

Introduction to Aerial Robotics



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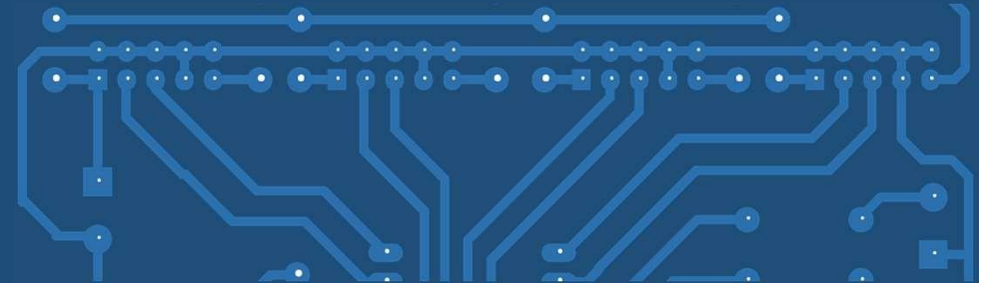


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EEEE4138 Aerial Robotics

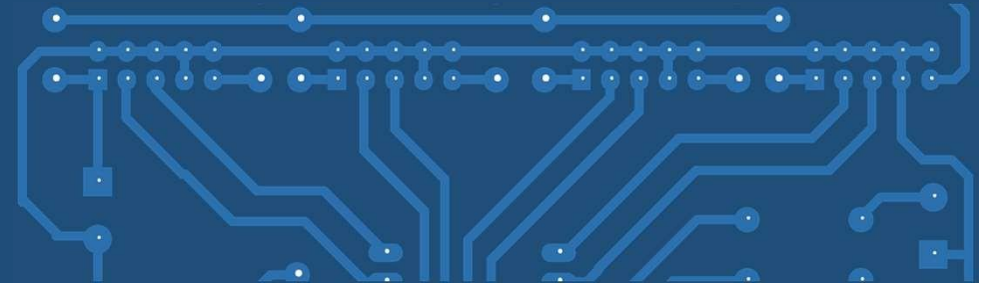
Learning Objectives



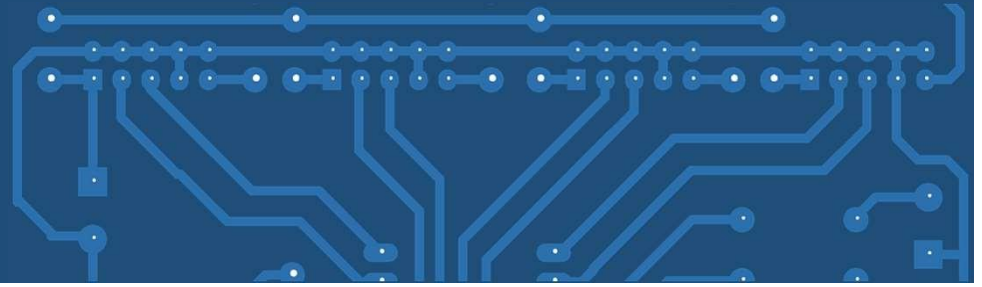
The aim of this module is to introduce students to the fundamental concepts of aerial robotics and to teach them how to model and control these flying robots in a safe and autonomous way. Additionally, this course prepares them to adapt with current and future aerospace technology industries.

EEEE4138 Aerial Robotics

Learning Contents

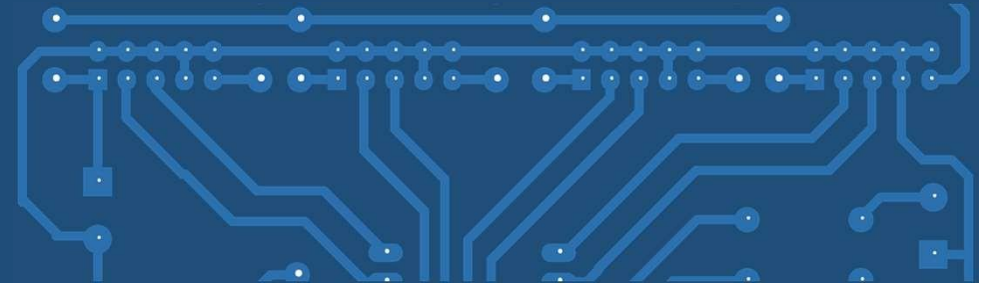


- ☐ Introduction + History
- ☐ Reference Frames
- ☐ Kinematics and Dynamics
- ☐ Forces, Moments and Actuation
- ☐ Flight Control
- ☐ Trajectory Generation
- ☐ Sensors
- ☐ Sensing and Estimation



On successful completion of this module, students will be able to:

- ☐ LO1 – Demonstrate basic knowledge of the physical principles behind the flight dynamics of aerial robotics and derive the mathematical model of small unmanned aerial vehicle.
- ☐ LO2 – Select and employ suitable techniques to sense and/or estimate the variables of these aerial robots.
- ☐ LO3 – Design high-performance flight controllers and develop algorithms for autonomous motion planning for aerial robotics.
- ☐ LO4 – Use modern software tools (like Matlab/Simulink) for analysis and control design of aerial robots.



☐ Teaching schedule:

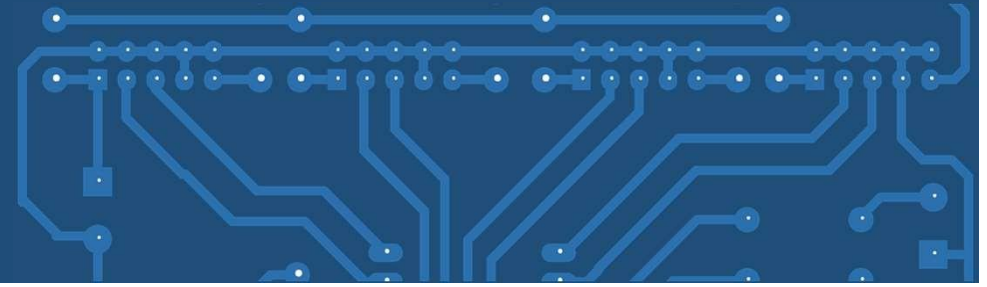
- ☐ 12 Lectures (2 hours per week) - Wednesdays 9am-11am@F4A11
- ☐ 12 Tutorials (1 hour per week) – Tuesdays 1pm-2pm@F3C03
- ☐ 5 laboratory sessions, 3 hours per session – Mondays 2pm-5pm@NA16a

☐ Exam, 60.0%

- ☐ The exam will cover all topics and LOs except for LO4. The duration will be 2 hours.

☐ Coursework, 40.0%

- ☐ MATLAB Simulink based simulations + report



Recommended Reading:

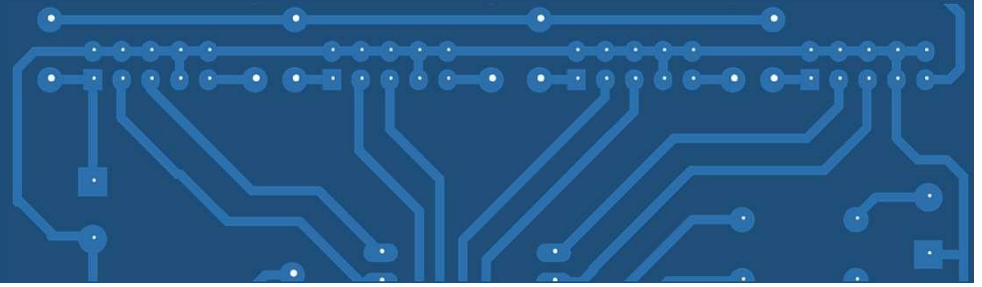
Randal W. Beard; Timothy W. McLain, *Small unmanned aircraft: theory and practice*, Princeton University Press, 2012.

Saeed B. Niku, *Introduction to robotics: analysis, control, applications*, 3rd edition, Wiley, 2020.

Supplementary Reading:

Barton, J. D., *Fundamentals of small unmanned aircraft flight*, Johns Hopkins APL Technical Digest, vol. 31, no. 2, pp. 132–149, 2012.

Shinde, S. K., More, J. S. and Chaudhari, C. P., *Mastering Drone Technology with AI: A Comprehensive Guide to Drone Operations and Techniques*, BPB Publications, 2025.



Unmanned Aerial Vehicles

Unmanned Aerial Vehicle (UAV)

An Unmanned Aerial Vehicle (UAV) is an aircraft with no human pilot on board. Configurations include fixed-wing aircraft, helicopters, multirotors and hybrid VTOL aircraft. UAVs may be remotely piloted or operate autonomously.

Unmanned Aircraft System (UAS)

The complete system: the UAV, its ground control station, communication links and supporting equipment.

Example: A quadrotor is a UAV. The quadrotor together with its ground station and communication links forms a UAS.



UAV Operations : Visual Line of Sight

VLOS — Visual Line of Sight: The remote pilot maintains direct visual contact with the UAV to monitor its flight and surrounding hazards.

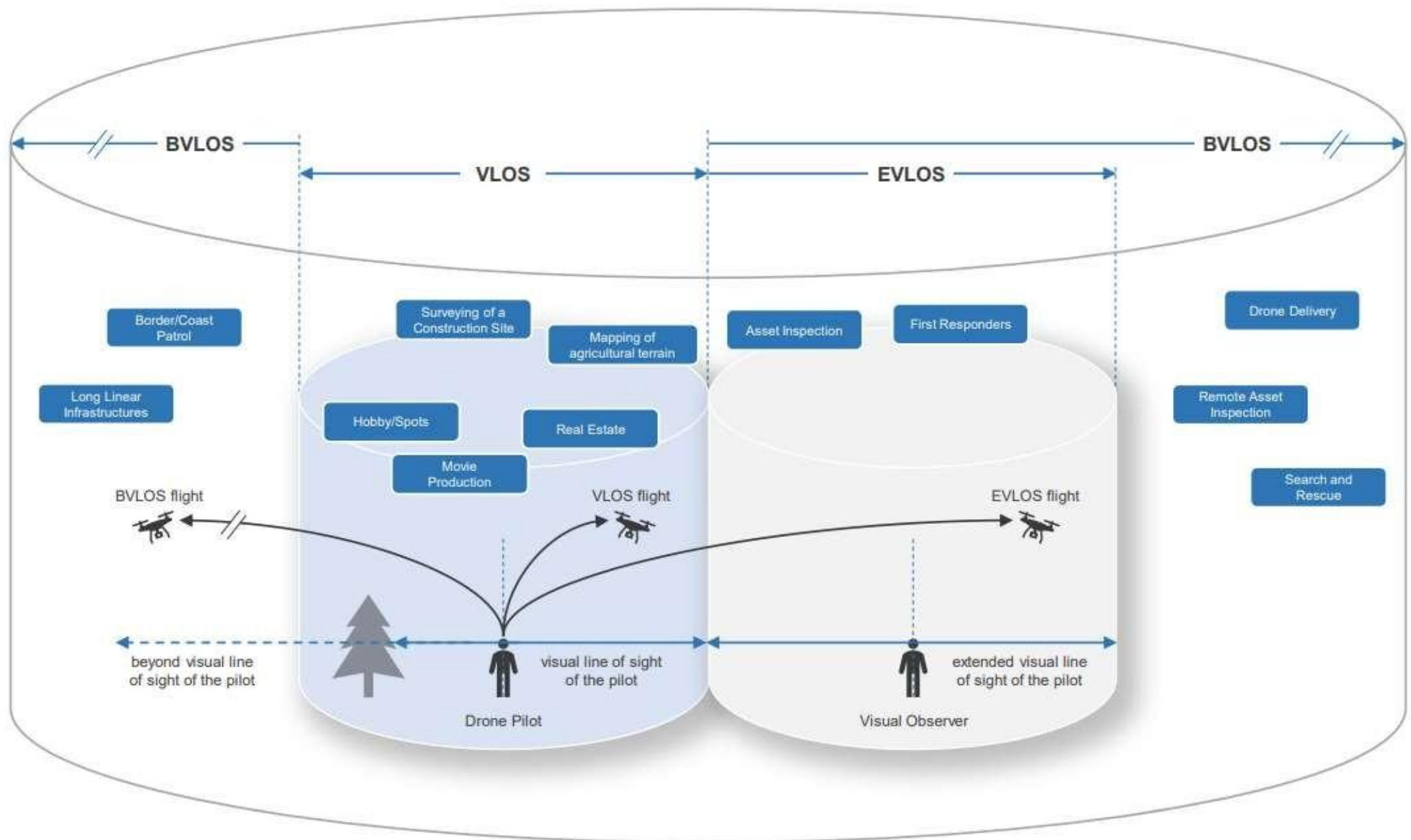
EVLOS — Extended Visual Line of Sight: Visual observers assist the remote pilot, extending operations beyond the pilot's own visual range.

BVLOS — Beyond Visual Line of Sight: Operations extend beyond direct visual observation, requiring other means of maintaining situational awareness and managing collision risks.

Regulatory context: In Malaysia, CAAM oversees UAS flight approvals. The operating category affects the applicable approval and safety requirements.

[CAAM — Unmanned Aircraft System guidance.](#)

VLOS, EVLOS & BVLOS Operation



UAV Structures

lighter than air

balloon



blimp



heavier than air

fixed wing



small UAV



MALE



HALE

hybrid



VTOL fixed wing



tilt wing



tilt engine



tilt platform

multi-rotor



tri-copter



quad-copter



hexa-copter



octo-copter



conventional



coaxial



nano



flettner

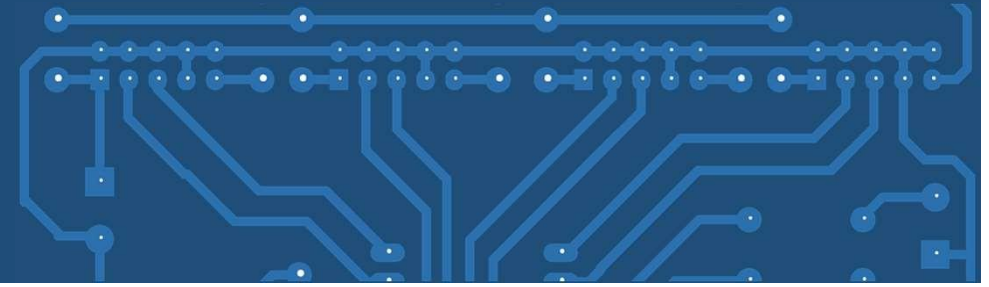
MALE: medium altitude long endurance (15.000 – 45.000 ft.), HALE: high altitude long endurance (>45.000 ft.)

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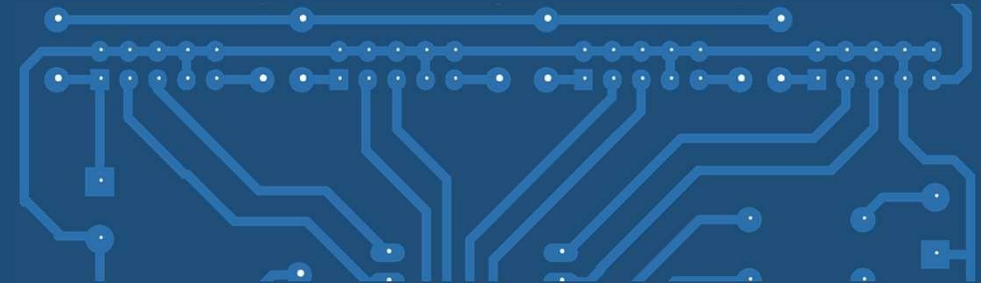
UAVs: Fixed Wings



- Generally efficient for forward flight and wide-area coverage
- Great for wide-area mapping
 - Cannot hover for stationary close-up inspection.
 - Problems doing low-level terrain following in hilly regions
- Requires take-off and landing area
 - Throw Launch
 - Catapult or Bungee
 - Belly Landings Common - Damage to airframe and payload
 - Parachute Landings - Weight, Complexity, Reliability, Wind Drift



- Leading simplicity and ease of use
 - Few moving parts
 - VTOL
 - Wide selection of Ready-to-Fly systems
- Short flight times
- Low speed
 - Low area coverage per battery charge
 - But easy to change batteries and go
- Does not scale well
 - Practical limits on size and duration



- Middle-range flight duration and speed
 - 30-60 minutes
 - 70-100 km/h
- Stable hover
- Ease of transport
 - Heavy payload capability compared to size
- Mechanical complexity and fragile
- Dangerous? Difficult?
 - Not more dangerous than comparable multirotor
- Gas Power
 - VTOL performance with multi-hour duration and high speed





- Combine range of airplane, with VTOL of Multirotor
- Reduced performance compared to airplane
 - Less range, lower speed
 - Less payload weight and volume





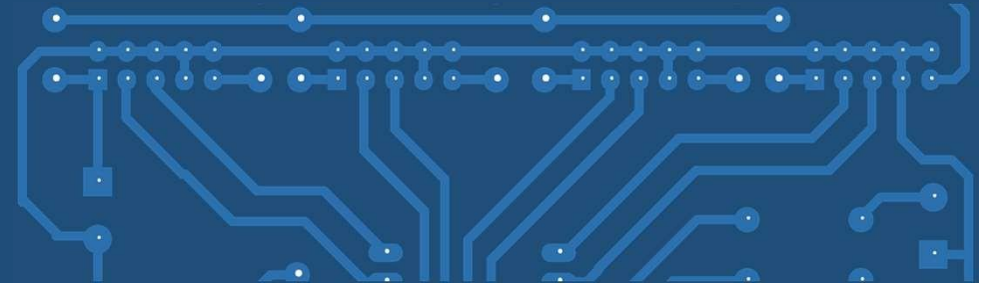
Mapping

Process of creating a diagrammatic representation (incl. 3D modelling) of a given area.



Inspection

Examination of a given object with the intent to find faults, errors, malfunctions or other specific phenomena that might affect the functioning of it



Surveying

Accurate measurement of a section of the earth's surface with the purpose to study or measure altitudes, angles, distances



Photography & Filming

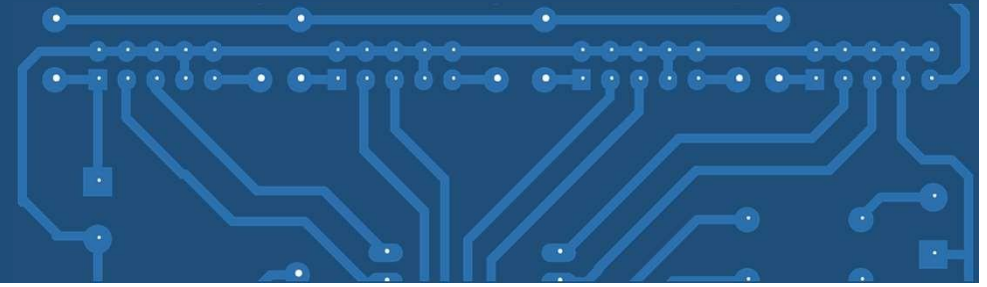
Aerial use of cameras to capture images and the production of aerial videos for all entertainment or advertising purposes.



Dispensing & Spraying

Aerial distribution of solids (e.g. fertiliser or seeds), or the process of spreading liquid substances (e.g. water, insecticide or fertiliser)





Monitoring

Closely observing a given object to check its progress or quality over a given period of time.



Delivery

Transportation of packages, food, medicines, or other goods either on demand or according to a given schedule.





Warehousing

The use of drones to assist the process of storing large quantities of goods so that they can be sold or used at a later date by managing the inventory.



Localization & Detection

Seeking and supplying the geographical coordinates of activities, persons or livestock, or detecting certain activities.

Others

Advertising, Broadcasting, Entertainment, Measuring & Sampling, Navigating, Tracking & Passenger Transport.

UAVs to the Rescue!

- Delivering medicine and blood through UAVs
- Resources:
 - Zipline: flyzipline.com



https://www.youtube.com/watch?v=jEbRVNxL44c&feature=emb_logo



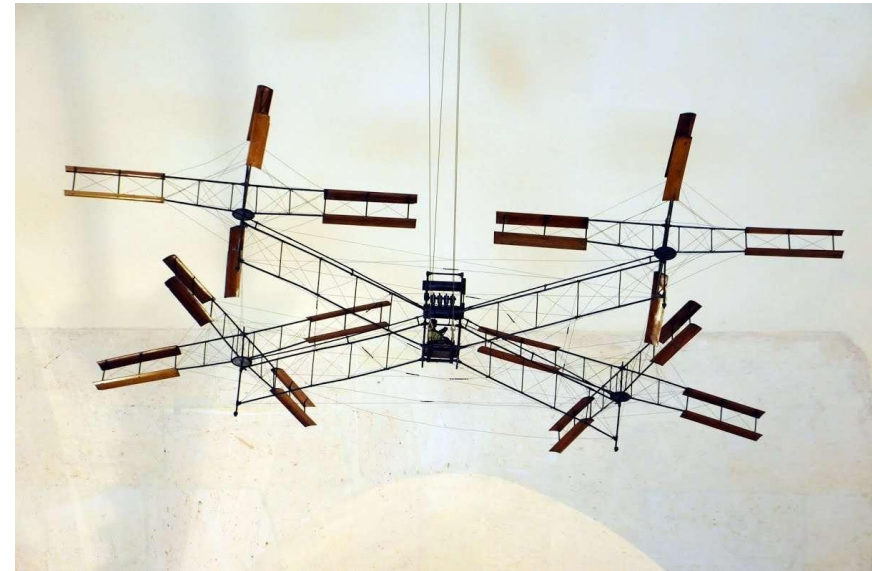
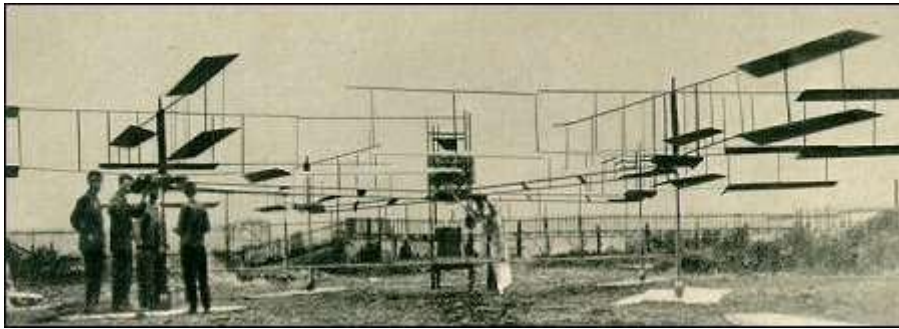
Orville Wright, Wilbur Wright, First Flight Military Airplane 1909

<https://www.youtube.com/watch?v=dtZ8MxuePno>



Quadrotor: Gyroplane No.1

Louis Breguet, Jacques Breguet and assisted by Professor Charles Richet, 1907



- Crew: **1**,
- Engine: gasoline engine rated at **29kW**,
- Rotor diameter: **8.0m**,
- Take-off weight: **578kg**,
- Empty weight: **340kg**



Hewitt-Sperry Automatic Airplane

Testing the Aerial Torpedo on
the Long Island Motor Parkway.
1917



What Enabled Modern Aerial Robots?

- ✓ Small, affordable inertial sensors for motion measurement.
- ✓ Embedded processors for rapid feedback control.
- ✓ Lightweight batteries, electric motors and motor controllers.
- ✓ Satellite navigation and wireless communication.
- ✓ Advances in estimation, control and motion-planning algorithms.



Activity : Which UAV would you choose?

Mission A : Building inspection

Inspect cracks on a university building façade. The aircraft must hover to capture detailed images. Take-off and landing space is limited.

Mission B : Agricultural mapping

Produce an aerial map of a large plantation. The aircraft must cover a wide area efficiently. Assume a suitable launch and recovery area is available, and hovering is unnecessary.

Discuss with a partner:

Which configuration would you choose for each mission: fixed-wing, helicopter, multicopter or hybrid VTOL?

Give two reasons for each choice.

Identify one limitation of your chosen aircraft.

